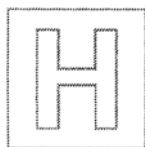


NAME: _____ CLASS: _____ INDEX: _____



CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 2

BIOLOGY

Paper 1 Multiple Choice

9648/01

2nd September 2014

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write and/or shade your name, NRIC / FIN number and HT group on the Answer Sheet in the spaces provided unless this has been done for you.

There are **40 MCQ** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **2B pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

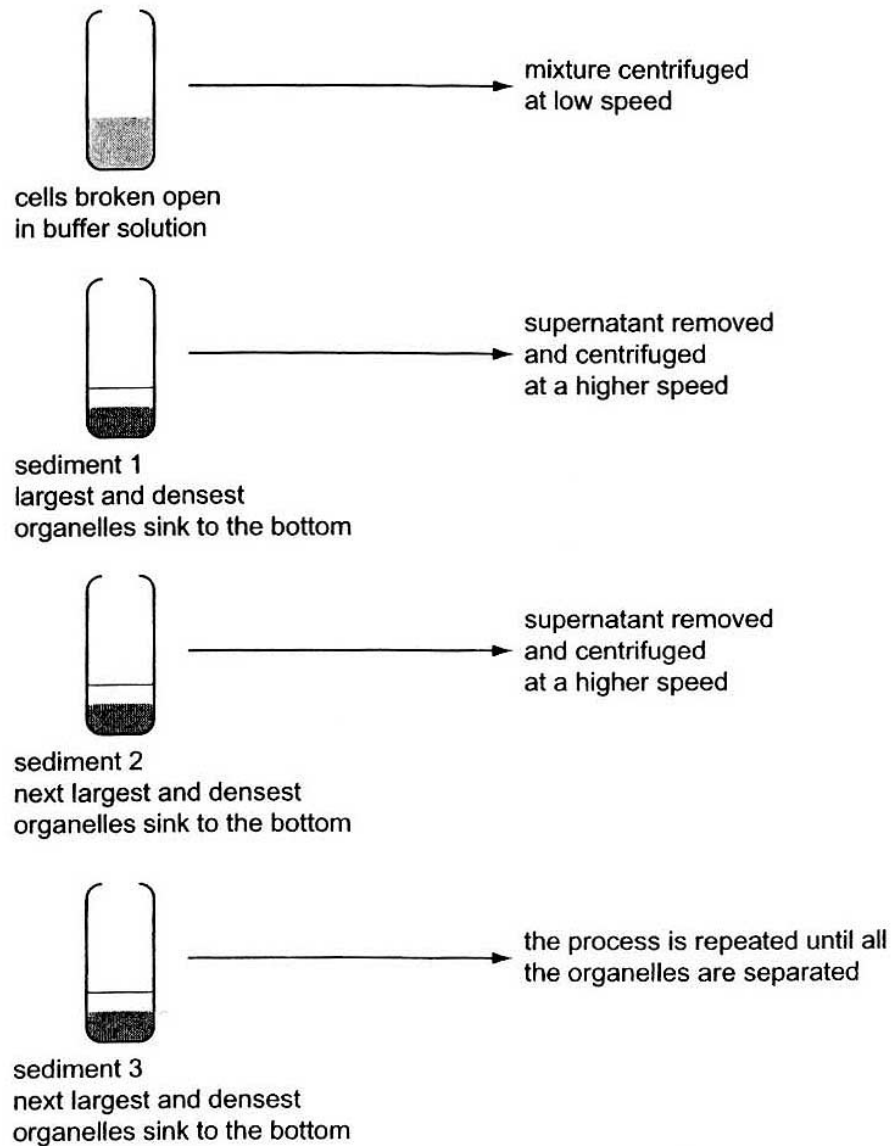
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Calculators may be used.

1. Fractionation is a process used to separate cell components according to their size and density.

The diagram below shows the main steps in fractionation of plant cells.



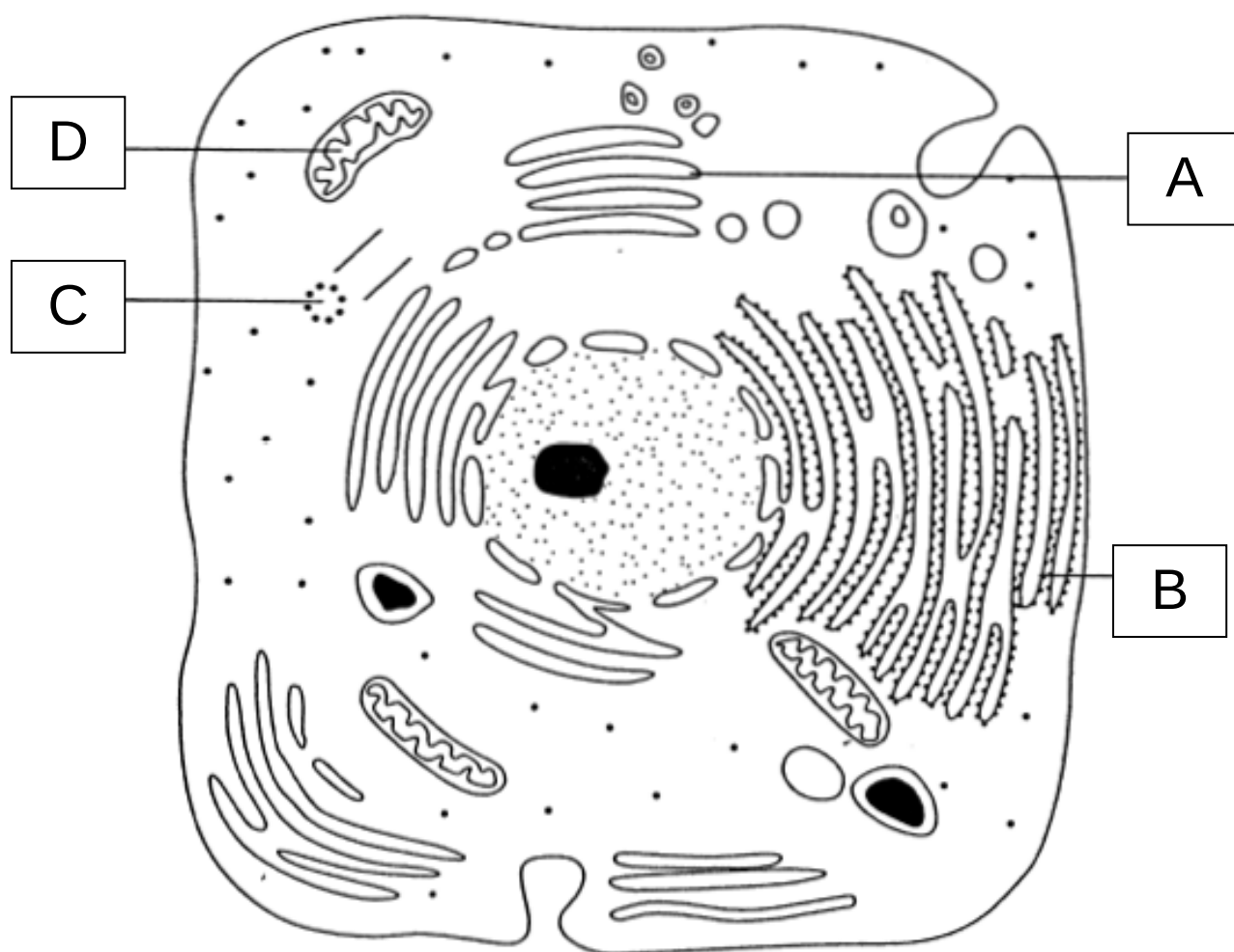
DCPIP and buffer solution were added to each sediment and the mixtures left in the light for fifteen minutes. Sediment 2 caused the oxidised DCPIP to be reduced.

Which organelles are present in sediment 2?

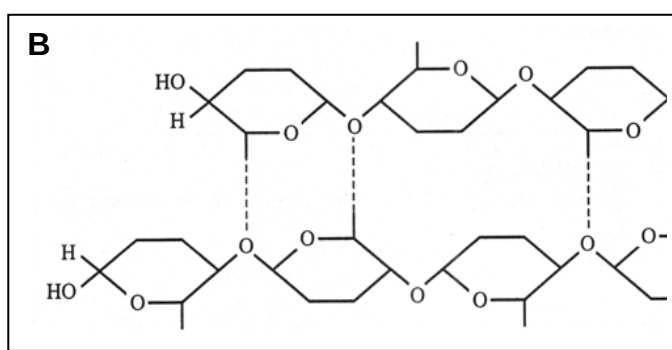
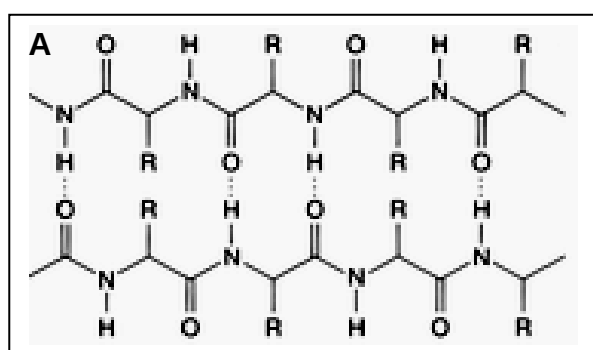
- A ribosomes
- B Mitochondria
- C Nuclei
- D Chloroplasts

2. The diagram shows the ultrastructure of a cell.

In which cell component is a newly synthesised protein modified by adding carbohydrate?



3. The following shows two molecules that perform structural functions.



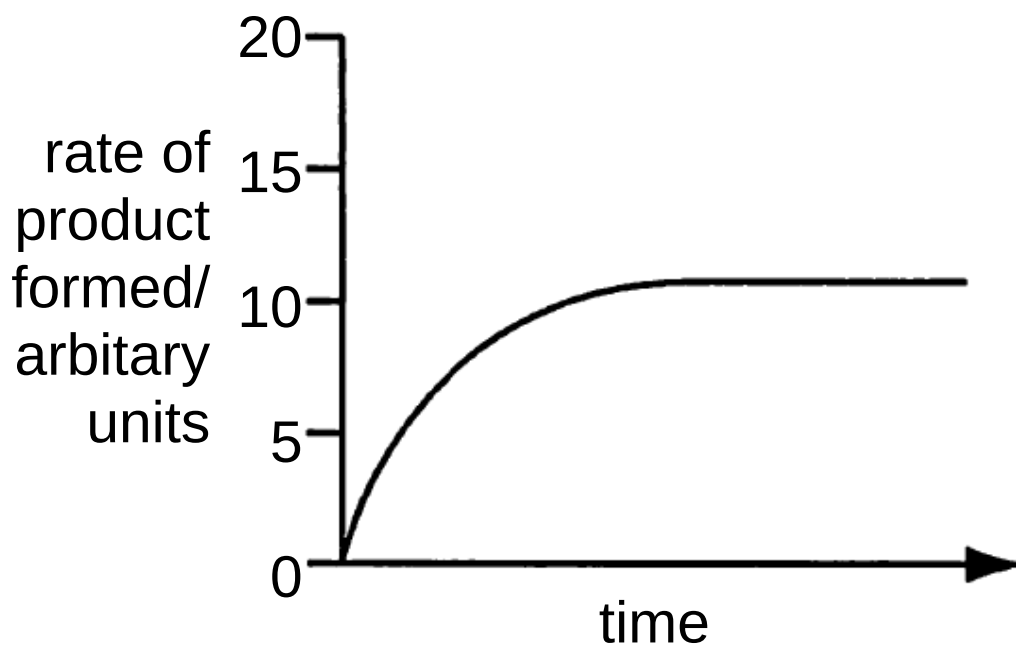
Which of the following statements does not describe a similarity between molecules A and B?

- A Hydrogen bonds form cross-linkages between neighbouring chains.
- B Monomers are joined together via peptide bonds.
- C Monomers can be extracted by hydrolysing the bonds within each chain.
- D Covalent bonds are present in both molecules.

4. The **R** group of the amino acid *serine* is $-\text{CH}_2-\text{OH}$. The **R** group of the amino acid *alanine* is $-\text{CH}_3$.

Where would you expect to find these amino acids in a globular protein in aqueous solution?

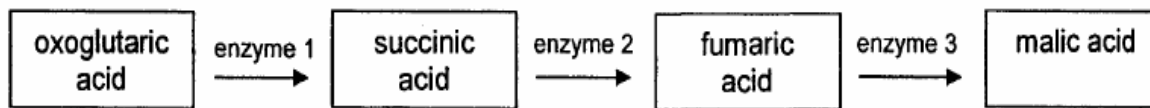
- A Serine would be in the interior, and alanine would be on the exterior of the globular protein.
 - B Alanine would be in the interior, and serine would be on the exterior of the globular protein.
 - C Both serine and alanine would be in the interior and on the exterior of the globular protein
 - D Both serine and alanine would be in the interior of the globular protein.
5. The graph shows the amount of product formed by a standard concentration of enzyme and a standard concentration of substrate at a temperature of 15°C .



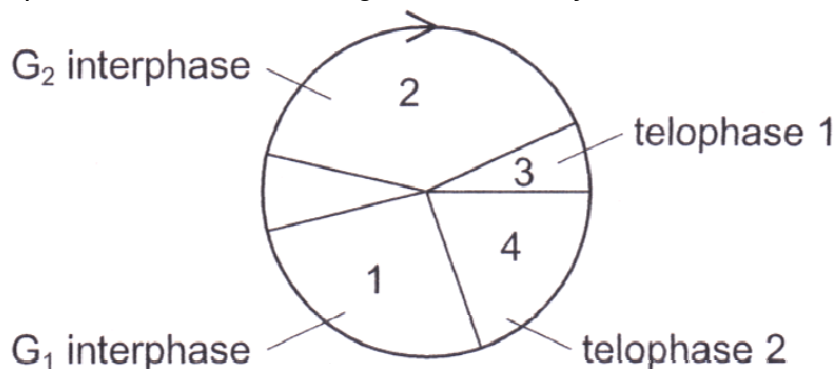
What would the rate be for an experimental temperature of 25°C

- A 5
- B 10
- C 15
- D 20

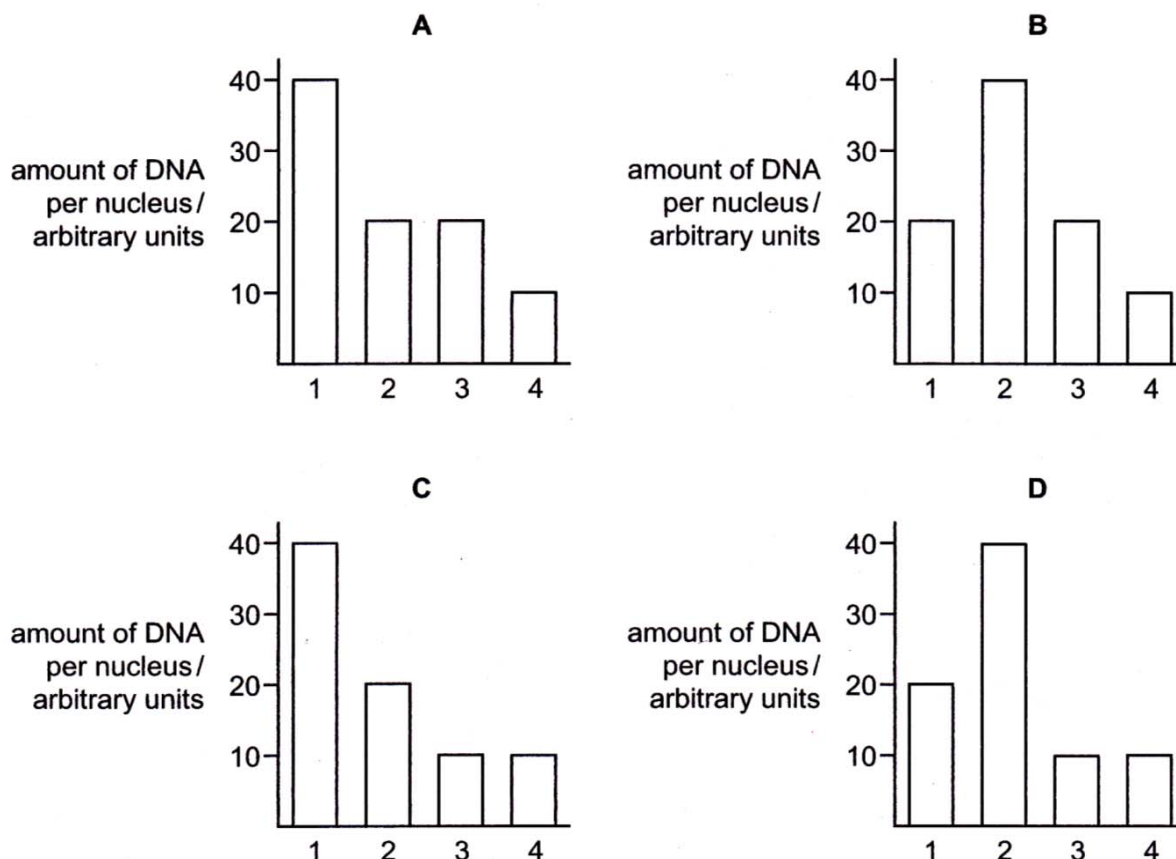
6. The four acids shown below form part of an enzyme-catalysed sequence of reactions. The addition of malonic acid results in no change in the concentration of oxoglutaric acid, an accumulation of succinic acid, and a near absence of both fumaric acid and malic acid. Further addition of fumaric acid results in the formation of malic acid. What does this information indicate about malonic acid?



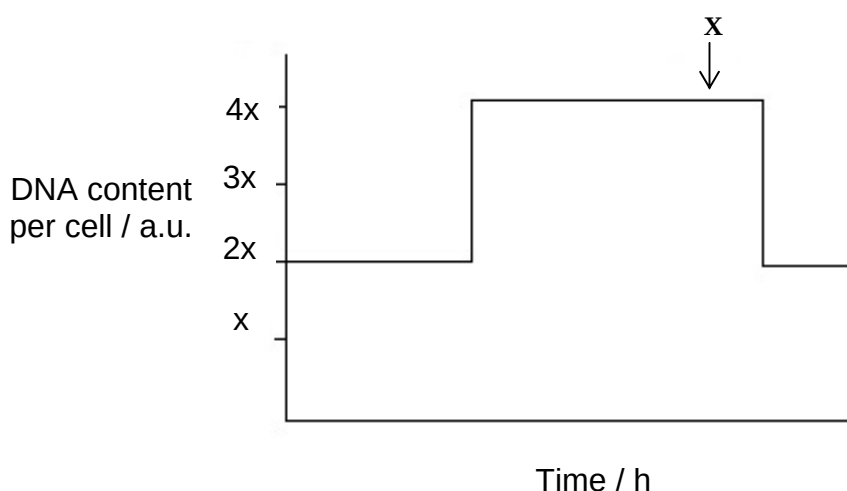
- A It is an inhibitor of enzyme 1.
 B It is an inhibitor of enzyme 2.
 C It reacts with fumaric acid.
 D It catalyses the formation of succinic acid.
7. The amount of DNA per nucleus in a cell during a meiotic cell cycle, as shown below, was measured.



Which bar graph correctly represents the variation in the DNA content?



8. The graph represents the changes in the quantity of DNA present in a cheek cell. The cell has 18 chromosomes at prophase.

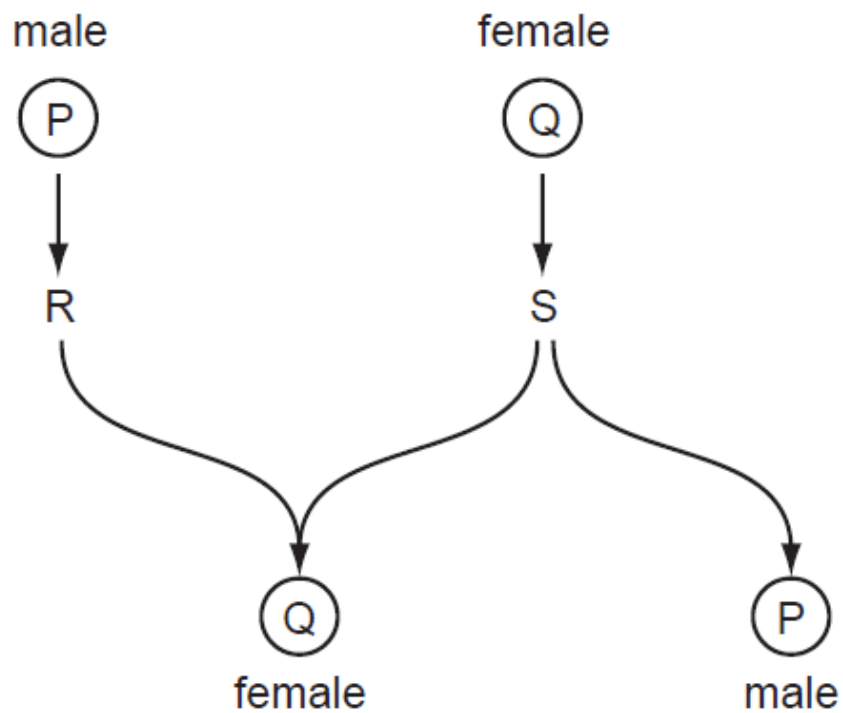


Which of the following combinations is correct at stage X?

	Number of chromosomes	Number of centromeres	Process occurring at stage X
A	18	18	The daughter chromosomes start to loosen and uncoil.
B	36	36	The sister chromatids have separated and formed daughter chromosomes as they move towards the poles.
C	18	36	The daughter chromosomes start to loosen and uncoil.
D	36	18	The sister chromatids have separated and formed daughter chromosomes as they move towards the poles.

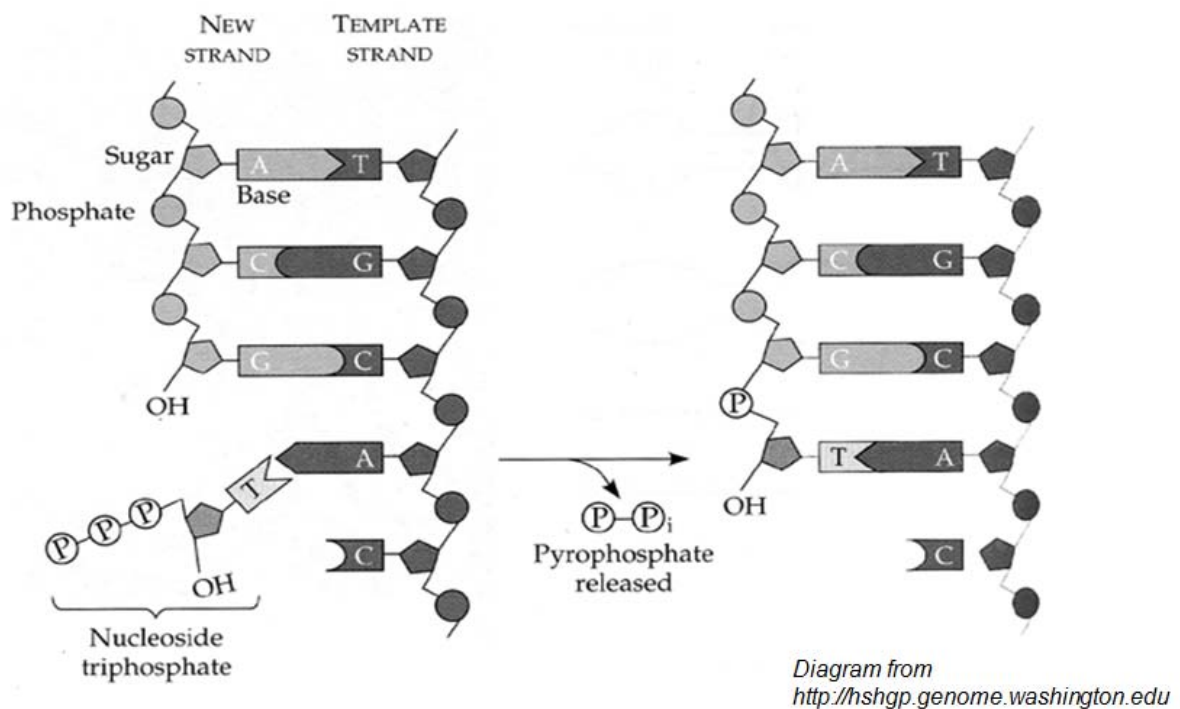
9. Sex determination in some insects such as bees and wasps is not controlled by sex chromosomes.

Using the diagram, which row in the table shows how sex is determined in these insects?



	P (ploidy)	Q (ploidy)	R (process)	S (process)
A	Haploid	Haploid	mitosis	mitosis
B	Haploid	Diploid	mitosis	meiosis
C	Diploid	Haploid	meiosis	meiosis
D	Diploid	Diploid	meiosis	Mitosis

10. The diagram below shows the synthesis of a new strand of DNA during interphase.

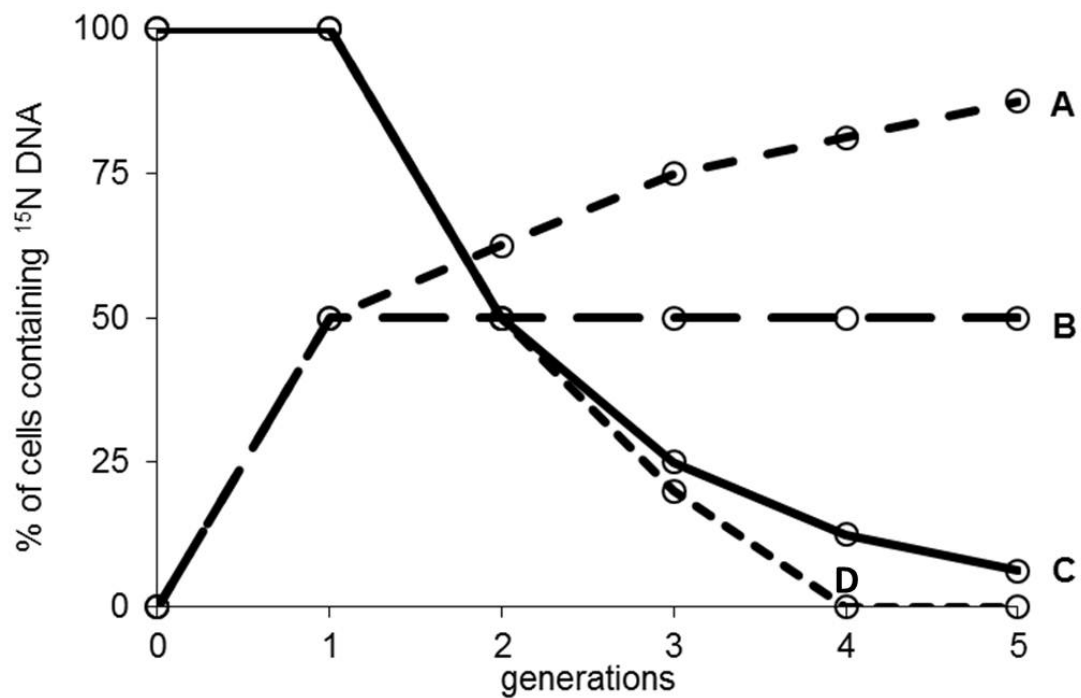


Which of the following shows the correct combination of bond(s) that need to be formed and the kind of reaction that is involved in order for the nucleotide to be added to the DNA chain?

	Bond(s) to be formed	Reaction(s) involved
A	Phosphodiester	Condensation and Dephosphorylation
B	Phosphodiester	Condensation and Hydrolysis
C	Phosphoester and Hydrogen	Condensation and Hydrolysis
D	Phosphoester and Hydrogen	Condensation

11. Bacteria were cultured in a medium containing heavy nitrogen (^{15}N) until all the DNA was labelled. These bacteria were then grown in a medium containing only normal nitrogen (^{14}N) for five generations. The percentage of cells containing ^{15}N in each generation was estimated.

Which curve provides evidence that DNA replication is semi-conservative?



12. Which of the following components is not directly required during translation?

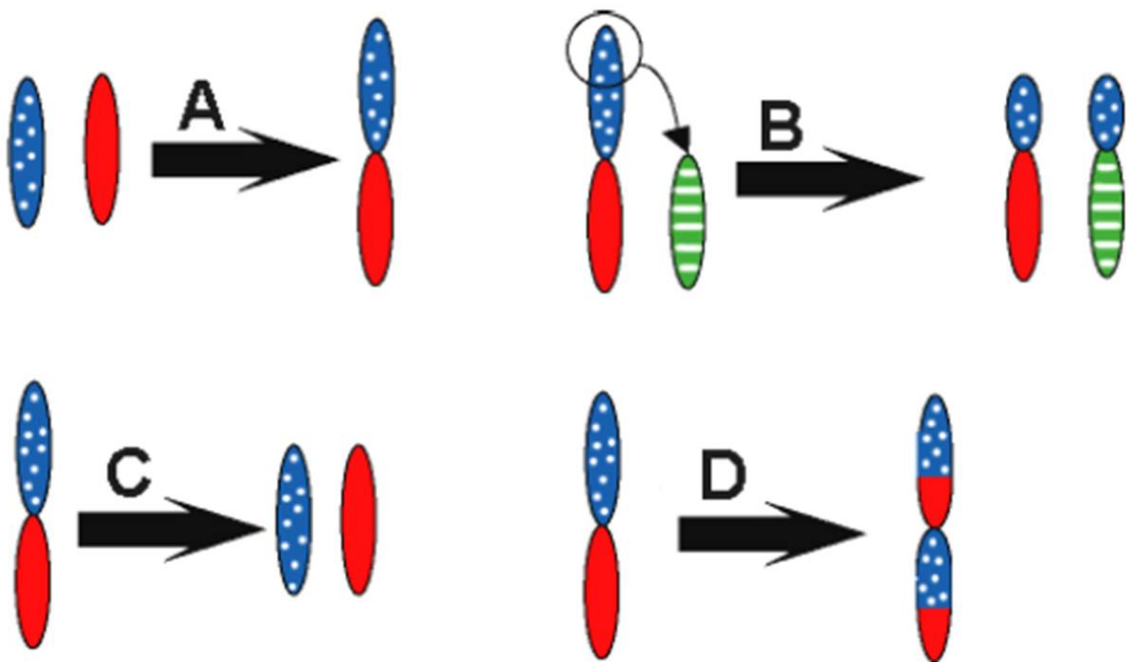
- A Aminoacyl-tRNA synthetase
- B Small ribosomal subunit
- C Peptidyl transferase
- D rRNA

13. Which of the following statement is true of the end-replication problem of linear DNA?

- 1 When a linear DNA molecule replicates, a gap is left at the 3' end of each new strand because DNA polymerase can only add nucleotides to a 3' end.
- 2 Telomerase prevents the end-replication problem from occurring.
- 3 Repeated rounds of replication produce shorter and shorter DNA molecules.
- 4 Prokaryotes do not have the end-replication problem.

- A** 1, 2 and 3
B 1 and 4
C 1, 3 and 4
D 3 and 4

14. Which of the following illustrates chromosomal inversion?



- 15.** Two structural proteins, J and K, are each encoded by different alleles of the same gene. The amino acid sequences of the two structural proteins differ between positions 91 and 95 of the polypeptides.

The amino acid sequences of enzymes J and K, and the corresponding DNA sequence of structural protein J from position 90 to position 97 of the polypeptides, are shown in the table below.

	←N-terminal end amino acid position C-terminal end →							
	90	91	92	93	94	95	96	97
DNA triplet codes for structural protein J	TTT	TCA	GGT	AGT	GAA	TTA	CGA	CGA
Amino acid sequence of structural protein J	lys	ser	pro	ser	leu	asn	ala	ala
Amino acid sequence of structural protein K	lys	val	his	his	leu	met	ala	ala

The actual mRNA codons for the amino acids in these positions for structural proteins J and K, are shown in the table below.

Amino acid	lys	ser	pro	leu	asn	ala	val	his	met
mRNA codon (s)	AAA	AGU UCA	CCA	CUU UUA	AAU	GCU	GUC	CAU CAC	AUG

What could account for the difference in amino acid sequence of structural proteins J and K?

- A** Frame shift mutations in the DNA codes at position 91 and position 94.
- B** A deletion in the DNA code at position 91 and an insertion into the DNA code at position 96.
- C** A single frame shift by deletion in the DNA code at position 91.
- D** A change in the sequences of the second and third nucleotides at positions 91 and 92 of the DNA codes and frame shifts at positions 93 and 95.

16. Genetic diversity is maintained by all of the following except?

- A** Mutation
- B** Transduction
- C** Transformation
- D** Binary fission

17. Repressible genes code for the synthesis of enzymes involved in anabolic pathways.

Which mechanism in bacteria can be used to control the activity of repressible genes by preventing RNA polymerase from binding to the operator?

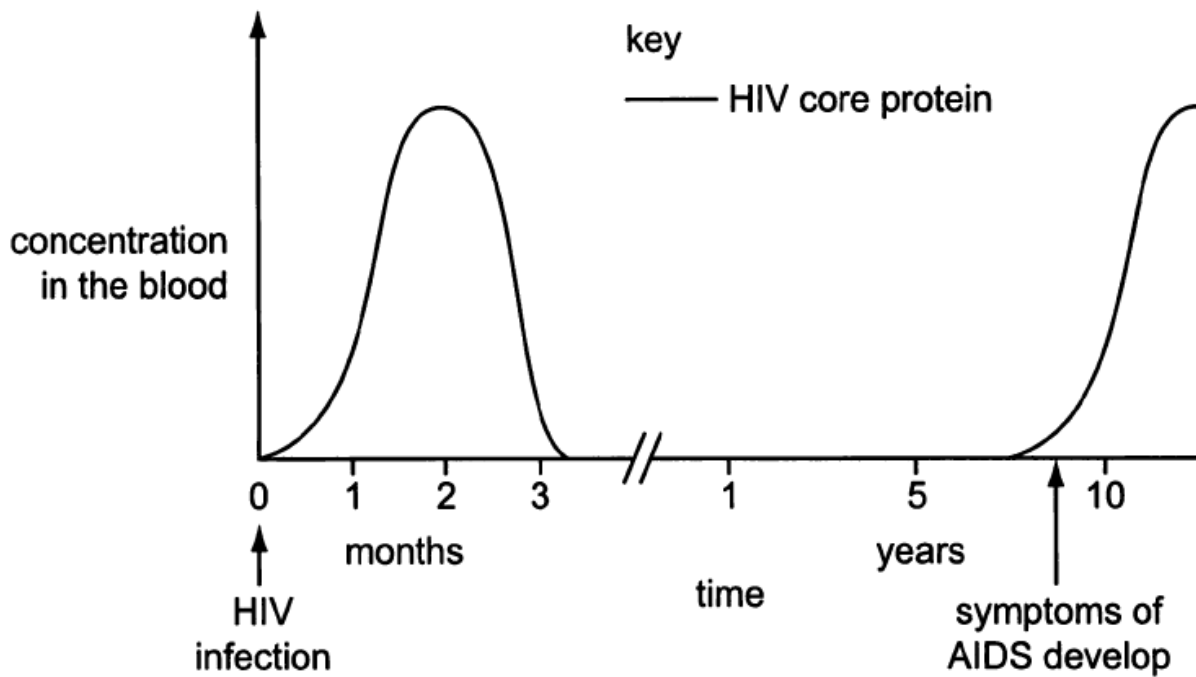
- A** Excess product of the anabolic pathway binds to the operator.
- B** Excess product of the anabolic pathway activates a repressor that binds to the operator.
- C** Excess substrate of the anabolic pathway binds to the operator.
- D** Excess substrate of the anabolic pathway binds to another protein that binds to the operator.

18. Which components of the influenza virus are likely to be targeted by drugs.

1. Haemagglutinin
2. Matrix protein
3. Neuraminidase
4. M2 protein ion channel
5. Lipid envelope
6. RNA-dependent RNA polymerases

- A** 1, 2 and 3
- B** 1, 2, 3 and 5
- C** 1, 3, 4, 5 and 6
- D** 1, 3, 4 and 6

19. The graph shows the changes in the core protein (capsid) concentration of the human immunodeficiency virus (HIV) in the blood stream after an HIV infection. The protein is detected using antibodies.

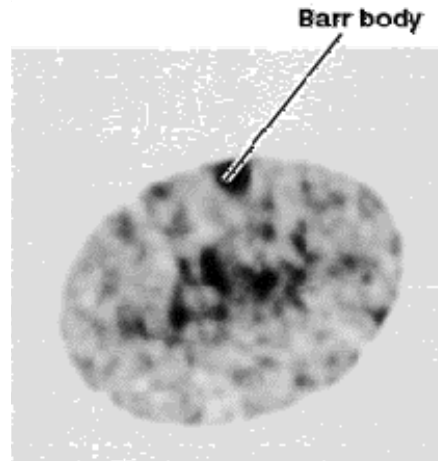


What explains the apparent absence of this protein between three months and seven years?

- A The patients' own antibodies destroy the core protein.
- B The virus enters the latent phase.
- C The virus integrates into the host cell genome and becomes a provirus.
- D The core protein attaches to a CD4 receptor so is not found in plasma.

20. The figure below shows a Barr body in cells of a normal human female. A Barr body is the inactivated X chromosome in a female somatic cell during embryogenesis. In human with more than one X chromosome, the number of Barr bodies is always one less than the total number of X chromosomes.

Which of the following statements about X chromosome inactivation is **false**?



- A The probability that a given X chromosome will become inactive is 50%.
 - B The inactivation of the X chromosome seems to involve methylation of cytosine on DNA
 - C The Barr body is a clump of euchromatin representing the inactivated X chromosome.
 - D An male with Klinefelter syndrome (genotype XXY) would have one Barr body.
21. Telomeres are non-coding DNA sequences usually found at the ends of chromosomes. In certain cells, telomere length can be increased by the action of the telomerase enzyme, coded for by the *TERC* gene.

Which of the following statements is/are **false**?

- 1 Telomerase is active only in stem cells and germ cells.
- 2 Telomere lengths shorten with every cell division due to incomplete replication of the lagging strand of DNA.
- 3 Telomerase adds nucleotide sequences to the 5' end of the DNA.
- 4 A gene mutation to *TERC* leads to reduced telomere length.

- A 1 only
- B 1 and 2 only
- C 1 and 3 only
- D 3 and 4 only

- 22.** Messenger RNA is isolated by passing homogenised tissue through a fractionating column. The column consists of short lengths of uracil nucleotides attached to a solid support medium. The mRNA that passes through the column cannot be translated as they break up into fragments. The mRNAs that attach to the column can be separated by application of sodium hydroxide.

Which statement is true of the mRNA in this experiment?

- (i) Complementary base-pairing occurs between adenine and uracil bases in the column.
- (ii) The mRNA that pass through the fractionating column probably lack poly A tails.
- (iii) The mRNA that attach to the fractionating column can be translated.
- (iv) The mRNA that attach to the fractionating column probably lack poly A tails.

- A** (i) and (iv)
- B** (i), (ii) and (iii)
- C** (ii) and (iv)
- D** All of the above.

- 23.** Which of the following is not a requirement for RNA splicing?

- A** Spliceosome
- B** ATP
- C** RNA polymerase
- D** Splice site

- 24.** In *E. coli* bacteria, regulation of gene expression involving the catabolite activator protein (CAP), is actually a type of positive regulation because

- A** cAMP-CAP helps RNA polymerase to increase rate of transcription.
- B** CAP increases the production of cAMP.
- C** glucose binds to CAP and activates it.
- D** glucose stimulates the production of cAMP.

- 25.** A bacterium produces a normal protein with the following amino acid sequence:

Met – Val – His – Lys – Arg – Thr – Leu – Val

After irradiation, a mutant strain is produced that synthesises a mutant protein from the same coding region on DNA with the following sequence:

Met – Val – His – Lys – Glu – Pro

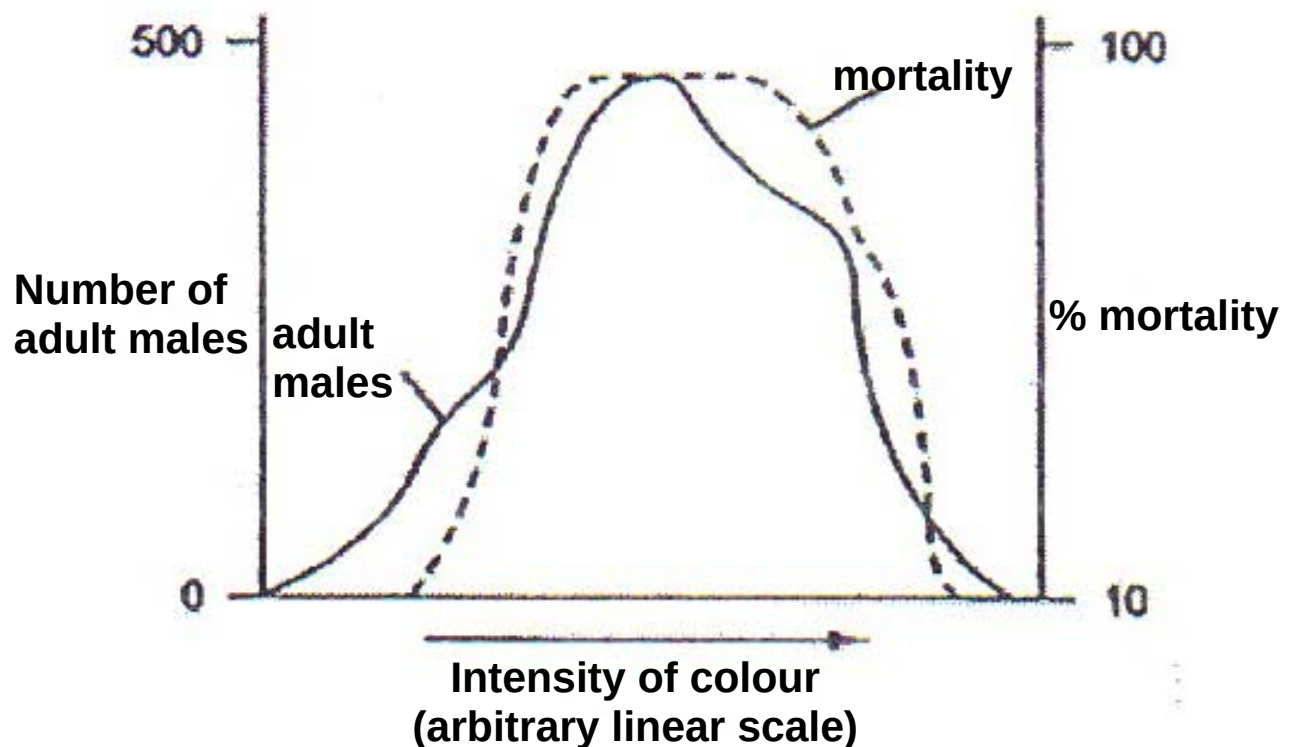
The mRNA codons for some amino acids are shown as follows:

Arg	Glu	Leu	Pro	Thr	Val
AGA	GAA	UUA	CCU	ACC	GUU
	GAG		CCC	ACG	GUA
				ACA	GUG

Which of the following mutations have occurred in the template DNA strand encoding the protein?

- A** Substitution of T by A at the 13th nucleotide position.
- B** Substitution of A by T at the 20th nucleotide position.
- C** Insertion of C at the 13th nucleotide position.
- D** Deletion of T at the 13th nucleotide position.

26. The graph below shows data on a population of a species of moth which shows considerable variation in colour intensity



Which conclusion can be made from this graph?

- A Colour variation is environmentally induced.
- B Colour variation is genetically determined.
- C Extreme forms are favoured by natural selection.
- D The species shows discontinuous variation with respect to colour.

27. Student X, looked after a plant. Another student, Y, looked after another plant of the same species. Each student followed the same instructions to set up apparatus to take cuttings from their plant and grow the cuttings next to the plant from which the cutting had been taken.

Fig. 27 shows the results after one week.

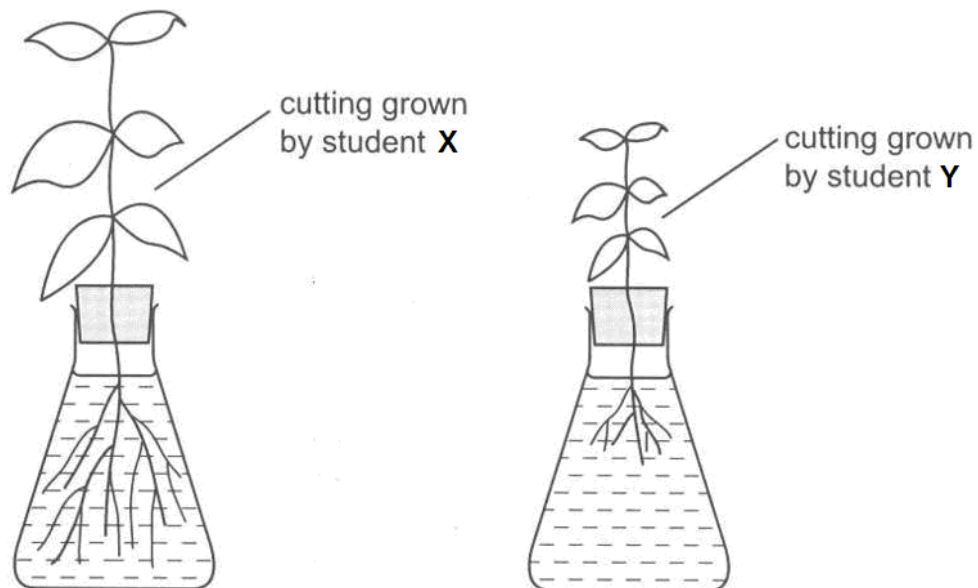


Fig. 27

Which factors could have caused the different appearance of the two cuttings?

- 1 phenotypic variation due to environment
- 2 genetically different cuttings
- 3 mutation due to environment
- 4 genotypic variation due to environment

A 1 and 2

B 2 and 3

C 2 and 4

D 1, 3 and 4

28. A yellow flower, green-stemmed plant with the genotype **YYrr** was crossed with a white flower, red-stemmed plant with the genotype **yyRR**. The F_1 plants were allowed to self fertilise, A chi-square test was carried out on the results obtained for the F_2 generation. Part of the table of values for chi-square is shown.

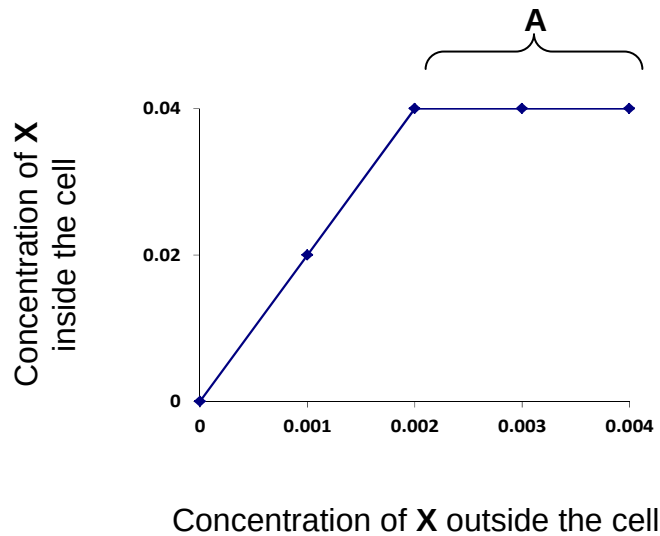
degrees of freedom	p = 0.5	p = 0.1	p = 0.05	p = 0.01	p = 0.001
1	0.46	2.71	3.84	6.64	10.83
2	1.39	4.61	5.99	9.21	13.82
3	2.37	6.25	7.82	11.34	16.27
4	3.37	7.78	9.49	13.28	18.46
5	4.35	9.24	11.07	15.09	20.52

The value of chi-square in this investigation was 10.6.

What is the probability of this value and do the results fit the expected ratio?

	<u>Probability</u>	<u>Results fit expected ratio</u>
A	Between 0.01 and 0.05	No
B	Between 0.01 and 0.05	Yes
C	Between 0.05 and 0.1	Yes
D	Between 0.1 and 0.5	No

29. Substance **X** is actively transported into cells. Equal-sized samples of cells were placed in media containing different concentrations of **X** for an hour. The intracellular concentration of **X** was then measured. All other metabolic conditions were maintained at the optimum level. The graph below shows the results.



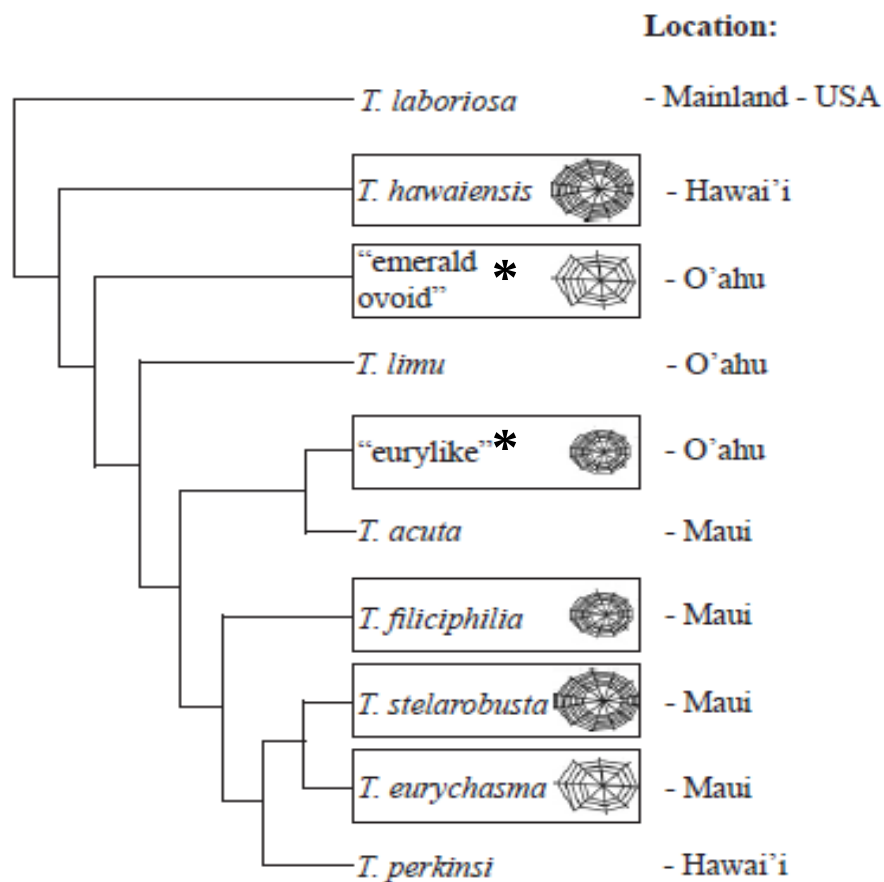
From the information given above, which one of the following would account for the region of the graph labelled A?

- A A respiratory inhibitor had been introduced.
- B The active transport carriers had been inactivated by a non-competitive inhibitor.
- C All the active transport carriers had been operating at their maximum rate.
- D As the internal concentration of **X** rose, more of the substance **X** was metabolised.

30. How does injection of insulin into diabetics lead to a lowering of blood glucose concentration?

- A by decreasing the permeability of cells to glucose
- B by increasing the excretion of glucose
- C by promoting the formation of the hormone glucagon
- D by promoting the synthesis of glycogen

31. The cladogram below shows the classification of a group of spiders found on the Hawaiian islands. An asterisk (*) indicates that this species of spider has yet to be assigned a scientific name.



[Source: T A Blackledge and R G Gillespie (November 2004),
Proceedings of the National Academy of Sciences, 101, (46), pages 16228–16233]

Based on information from the diagram above, deduce which of the following spiders are the most closely related species.

- A *T. filiciphilia* and "eurylike"
- B *T. hawaiiensis* and "emerald ovoid"
- C *T. stelarobusta* and *T. eurychasma*
- D *T. stelarobusta* and *T. filiciphilia*

32. Which effect of natural selection is likely to lead to speciation?

- A** Differences between populations are increased.
- B** Favorable genotypes are maintained in the population.
- C** Genetic diversity is reduced.
- D** Selection pressure on some alleles reduces reproductive success.

33. Which term considers only evolutionary relationships?

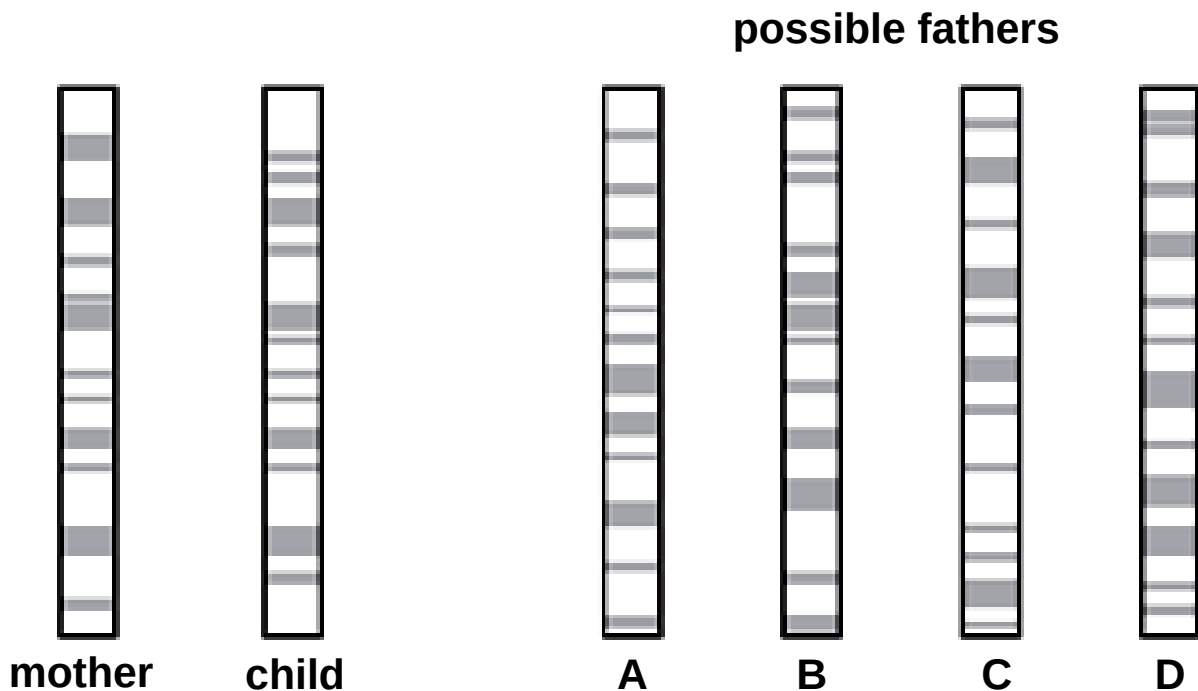
- A** classification
- B** natural selection
- C** Clade
- D** Taxonomy

34. Which of the following primers would allow for the copying of the single-stranded DNA sequence, 5' ATGCCTAGGTC 3' in the polymerase chain reaction (PCR)?

- A** 5' TACGG 3'
- B** 5' TCCAG 3'
- C** 5' GACCU 3'
- D** 5' GACCT 3'

35. Genetic profiling can be used to determine the paternity of a child. DNA from the mother and the child is cut into fragments, separated by electrophoresis and made visual using a stain. The diagram shows the genetic profiles of a mother and child, and four possible fathers.

Who is the father?



36. Which of the following is not a property of all stem cells?
- A Able to self-renew
 - B Able to proliferate continually
 - C Able to differentiate into specialised cells
 - D Totipotency
37. A patient suffering from cystic fibrosis underwent gene therapy using a viral-mediated delivery system. The treatment failed and you were asked to troubleshoot the problem.

Which of the following could not be the explanation behind the failure in the treatment?

- A Failure of the expressed protein to be folded into the correct conformation.
- B Integration of the target gene in the enhancer region.
- C Rejection of the vector by the host immune system.
- D The CFTR protein was not expressed in adequate amount.

38. The following steps are involved in the life cycle of a virus:

1. Release
2. Maturation
3. Adsorption
4. Replication
5. Penetration

Which is the correct order of the viral life cycle?

- A** 1, 3, 2, 4 & 5
- B** 3, 5, 2, 4, & 1
- C** 5, 3, 2, 1 & 4
- D** 3, 5, 4, 2, & 1

39. Which of the following statements could not explain why infection with *Agrobacterium tumefaciens* for genetic engineering of plant cells is preferred?

- A** *Agrobacterium tumefaciens* can efficiently transfer DNA into plant cells upon infection.
- B** *Agrobacterium tumefaciens* can harbour the Ti plasmid in which foreign genes can be inserted within it.
- C** Plant cell protoplasts may impede DNA entry via gene gun delivery.
- D** Plant cells have a cellulose cell wall which may impede DNA entry via transformation.

40. Normal tomato plants have a gene that codes for an enzyme that softens tomatoes as they ripen. The DNA sequence for this gene is as follows:

3' ... T T A G C C T T A ... 5',

Genetically engineered tomatoes ripen and soften more slowly because of the insertion of a transgene which reduces the amount of the softening enzyme produced by hybridizing with mRNA encoding the enzyme.

Which of the following would be the mRNA sequence encoded by this transgene?

- A** 5' – A U U C C G A U U – 3'
- B** 5' – A A T C G G A A T – 3'
- C** 5' – A A U C G G A A U – 3'
- D** 5' – U U A G C C U U A – 3'

END OF PAPER